Hospital Management System — Complete SQL Scripts

This PDF contains the complete SQL code from both uploaded files.

sql project.sql

```
create schema Hospital_Management_System;
use Hospital_Management_System;

CREATE TABLE Departments (
    department_id INT AUTO_INCREMENT PRIMARY KEY,
    department_name VARCHAR(100) NOT NULL UNIQUE,
    floor_no INT NOT NULL,
    head_doctor VARCHAR(100)
);

CREATE TABLE Patients (
    patient_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50),
    gender ENUM('Male','Female','Other') NOT NULL,
    date_of_birth DATE NOT NULL,
    blood_group VARCHAR(5),
    phone VARCHAR(15) UNIQUE,
    email VARCHAR(100) UNIQUE,
    address VARCHAR(255),
    city VARCHAR(50),
    admit_date DATE,
    discharge_date DATE
);


CREATE TABLE Doctors (
    doctor_id INT AUTO_INCREMENT PRIMARY KEY,
    doctor_name VARCHAR(100) NOT NULL,
    specialization VARCHAR(100),
    qualification VARCHAR(100),
    experience_years INT,
    phone VARCHAR(15) UNIQUE,
    email VARCHAR(100) UNIQUE,
    department_id INT NOT NULL,
    joining_date DATE,
    FOREIGN KEY (department_id)
    REFERENCES Departments(department_id)
);

CREATE TABLE Rooms (
    room_id INT AUTO_INCREMENT PRIMARY KEY,
    room_number VARCHAR(10) UNIQUE NOT NULL,
    room_type ENUM('General','Semi Private','Private','ICU') NOT NULL,
    floor INT,
    daily_charge DECIMAL(10,2),
    availability ENUM('Available','Occupied') DEFAULT 'Available'
);

CREATE TABLE Appointments (
    appointment_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    appointment_date DATE NOT NULL,
    appointment_time TIME,
    visit_type ENUM('New','Follow-up'),
    status ENUM('Completed','Cancelled','Pending'),
    consultation_fee DECIMAL(10,2),

    FOREIGN KEY(patient_id)
    REFERENCES Patients(patient_id),

    FOREIGN KEY(doctor_id)
    REFERENCES Doctors(doctor_id)
);

CREATE TABLE Admissions (
    admission_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    room_id INT NOT NULL,
    admission_date DATE,
    discharge_date DATE,
    reason VARCHAR(255),

    FOREIGN KEY(patient_id)
    REFERENCES Patients(patient_id),

    FOREIGN KEY(doctor_id)
    REFERENCES Doctors(doctor_id),

    FOREIGN KEY(room_id)
    REFERENCES Rooms(room_id)
);
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CREATE TABLE Medicines (
    medicine_id INT AUTO_INCREMENT PRIMARY KEY,
    medicine_name VARCHAR(100) NOT NULL,
    category VARCHAR(50),
    manufacturer VARCHAR(100),
    unit_price DECIMAL(10,2),
    stock_quantity INT,
    expiry_date DATE
);

CREATE TABLE Prescriptions (
    prescription_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    medicine_id INT NOT NULL,
    dosage VARCHAR(100),
    days INT,
    prescription_date DATE,

    FOREIGN KEY(patient_id)
    REFERENCES Patients(patient_id),

    FOREIGN KEY(doctor_id)
    REFERENCES Doctors(doctor_id),

    FOREIGN KEY(medicine_id)
    REFERENCES Medicines(medicine_id)
);

CREATE TABLE Bills (
    bill_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT NOT NULL,
    admission_id INT NOT NULL,
    doctor_fee DECIMAL(10,2),
    room_charge DECIMAL(10,2),
    lab_charge DECIMAL(10,2),
    medicine_charge DECIMAL(10,2),
    other_charge DECIMAL(10,2),
    discount DECIMAL(10,2),
    total_amount DECIMAL(10,2),
    payment_status ENUM('Paid','Pending') DEFAULT 'Pending',
    bill_date DATE,

    FOREIGN KEY(patient_id)
    REFERENCES Patients(patient_id),

    FOREIGN KEY(admission_id)
    REFERENCES Admissions(admission_id)
);

CREATE TABLE Salary (
    salary_id INT AUTO_INCREMENT PRIMARY KEY,
    doctor_id INT NOT NULL,
    basic_salary DECIMAL(12,2),
    bonus DECIMAL(10,2),
    deduction DECIMAL(10,2),
    net_salary DECIMAL(12,2),
    salary_month VARCHAR(20),
    salary_year INT,
    payment_date DATE,

    FOREIGN KEY(doctor_id)
    REFERENCES Doctors(doctor_id)
);

INSERT INTO Departments (department_id, department_name, floor_no, head_doctor) VALUES
(1, 'Cardiology', 1, 'Dr. Rajesh Kumar'),
(2, 'Neurology', 2, 'Dr. Priya Sharma'),
(3, 'Orthopedics', 3, 'Dr. Amit Verma'),
(4, 'Pediatrics', 2, 'Dr. Sneha Reddy'),
(5, 'Dermatology', 4, 'Dr. Kiran Patel'),
(6, 'ENT', 3, 'Dr. Suresh Rao'),
(7, 'General Medicine', 1, 'Dr. Anjali Gupta'),
(8, 'Gynecology', 2, 'Dr. Kavitha Nair'),
(9, 'Urology', 4, 'Dr. Vikram Singh'),
(10, 'Oncology', 5, 'Dr. Meena Joshi'),
(11, 'Nephrology', 5, 'Dr. Arjun Reddy'),
(12, 'Psychiatry', 6, 'Dr. Divya Menon'),
(13, 'Pulmonology', 4, 'Dr. Rohit Agarwal'),
(14, 'Gastroenterology', 3, 'Dr. Nikhil Babu'),
(15, 'Endocrinology', 5, 'Dr. Lakshmi Devi'),
(16, 'Ophthalmology', 2, 'Dr. Ravi Teja'),
(17, 'Dentistry', 1, 'Dr. Pooja Jain'),
(18, 'Emergency', 1, 'Dr. Harish Kumar'),
(19, 'Radiology', 6, 'Dr. Sunil Reddy'),
(20, 'Physiotherapy', 2, 'Dr. Ashok Varma');


INSERT INTO Doctors
(doctor_id, doctor_name, specialization, qualification, experience_years, phone, email, department_id, joining_date)
VALUES
(1, 'Dr. Rajesh Kumar', 'Cardiologist', 'MBBS, MD Cardiology', 15, '9876500001', 'rajesh.kumar@hospital.com', 1, '2015-06-10'),

(2, 'Dr. Priya Sharma', 'Neurologist', 'MBBS, DM Neurology', 12, '9876500002', 'priya.sharma@hospital.com', 2, '2016-03-15'),

(3, 'Dr. Amit Verma', 'Orthopedic Surgeon', 'MBBS, MS Orthopedics', 10, '9876500003', 'amit.verma@hospital.com', 3, '2018-07-21'),
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(4, 'Dr. Sneha Reddy', 'Pediatrician', 'MBBS, MD Pediatrics', 8, '9876500004', 'sneha.reddy@hospital.com', 4, '2019-01-18'),

(5, 'Dr. Kiran Patel', 'Dermatologist', 'MBBS, MD Dermatology', 11, '9876500005', 'kiran.patel@hospital.com', 5, '2017-09-12'),

(6, 'Dr. Suresh Rao', 'ENT Specialist', 'MBBS, MS ENT', 14, '9876500006', 'suresh.rao@hospital.com', 6, '2014-04-28'),

(7, 'Dr. Anjali Gupta', 'Physician', 'MBBS, MD General Medicine', 13, '9876500007', 'anjali.gupta@hospital.com', 7, '2015-12-05'),

(8, 'Dr. Kavitha Nair', 'Gynecologist', 'MBBS, MS Gynecology', 9, '9876500008', 'kavitha.nair@hospital.com', 8, '2020-02-14'),

(9, 'Dr. Vikram Singh', 'Urologist', 'MBBS, MCh Urology', 16, '9876500009', 'vikram.singh@hospital.com', 9, '2013-10-08'),

(10, 'Dr. Meena Joshi', 'Oncologist', 'MBBS, DM Oncology', 18, '9876500010', 'meena.joshi@hospital.com', 10, '2012-05-16'),

(11, 'Dr. Arjun Reddy', 'Nephrologist', 'MBBS, DM Nephrology', 9, '9876500011', 'arjun.reddy@hospital.com', 11, '2019-08-22'),

(12, 'Dr. Divya Menon', 'Psychiatrist', 'MBBS, MD Psychiatry', 7, '9876500012', 'divya.menon@hospital.com', 12, '2021-06-11'),

(13, 'Dr. Rohit Agarwal', 'Pulmonologist', 'MBBS, DM Pulmonology', 12, '9876500013', 'rohit.agarwal@hospital.com', 13, '2016-11-30'),

(14, 'Dr. Nikhil Babu', 'Gastroenterologist', 'MBBS, DM Gastroenterology', 10, '9876500014', 'nikhil.babu@hospital.com', 14, '2018-09-17'),

(15, 'Dr. Lakshmi Devi', 'Endocrinologist', 'MBBS, DM Endocrinology', 15, '9876500015', 'lakshmi.devi@hospital.com', 15, '2014-07-09'),

(16, 'Dr. Ravi Teja', 'Ophthalmologist', 'MBBS, MS Ophthalmology', 11, '9876500016', 'ravi.teja@hospital.com', 16, '2017-03-03'),

(17, 'Dr. Pooja Jain', 'Dentist', 'BDS, MDS', 8, '9876500017', 'pooja.jain@hospital.com', 17, '2020-10-25'),

(18, 'Dr. Harish Kumar', 'Emergency Physician', 'MBBS, MD Emergency Medicine', 14, '9876500018', 'harish.kumar@hospital.com', 18, '2015-01-12'),

(19, 'Dr. Sunil Reddy', 'Radiologist', 'MBBS, MD Radiology', 13, '9876500019', 'sunil.reddy@hospital.com', 19, '2016-08-19'),

(20, 'Dr. Ashok Varma', 'Physiotherapist', 'BPT, MPT', 10, '9876500020', 'ashok.varma@hospital.com', 20, '2018-12-01');


INSERT INTO Patients
(patient_id, first_name, last_name, gender, date_of_birth, blood_group, phone, email, address, city, admit_date, discharge_date)
VALUES
(1, 'Rahul', 'Sharma', 'Male', '1995-04-12', 'B+', '9876541001', 'rahul.sharma@gmail.com', 'Madhapur', 'Hyderabad', '2026-07-01', '2026-07-05'),

(2, 'Priya', 'Reddy', 'Female', '1998-09-18', 'A+', '9876541002', 'priya.reddy@gmail.com', 'Kukatpally', 'Hyderabad', '2026-07-02', '2026-07-04'

(3, 'Arjun', 'Kumar', 'Male', '1992-01-25', 'O+', '9876541003', 'arjun.kumar@gmail.com', 'Banjara Hills', 'Hyderabad', '2026-07-03', '2026-07-08

(4, 'Sneha', 'Patel', 'Female', '2000-11-10', 'AB+', '9876541004', 'sneha.patel@gmail.com', 'Ameerpet', 'Hyderabad', '2026-07-04', '2026-07-06')

(5, 'Kiran', 'Rao', 'Male', '1989-07-22', 'A-', '9876541005', 'kiran.rao@gmail.com', 'Miyapur', 'Hyderabad', '2026-07-05', '2026-07-10'),

(6, 'Anjali', 'Gupta', 'Female', '1996-05-16', 'B-', '9876541006', 'anjali.gupta@gmail.com', 'Gachibowli', 'Hyderabad', '2026-07-06', '2026-07-0

(7, 'Vikram', 'Singh', 'Male', '1993-08-28', 'O-', '9876541007', 'vikram.singh@gmail.com', 'Begumpet', 'Hyderabad', '2026-07-07', '2026-07-11'),

(8, 'Meena', 'Joshi', 'Female', '1997-02-14', 'AB-', '9876541008', 'meena.joshi@gmail.com', 'Secunderabad', 'Hyderabad', '2026-07-08', '2026-07-

(9, 'Rohit', 'Agarwal', 'Male', '1991-06-09', 'B+', '9876541009', 'rohit.agarwal@gmail.com', 'Kondapur', 'Hyderabad', '2026-07-09', '2026-07-13'

(10, 'Divya', 'Menon', 'Female', '1994-12-30', 'A+', '9876541010', 'divya.menon@gmail.com', 'LB Nagar', 'Hyderabad', '2026-07-10', '2026-07-14')
(11, 'Sunil', 'Reddy', 'Male', '1988-03-18', 'O+', '9876541011', 'sunil.reddy@gmail.com', 'Uppal', 'Hyderabad', '2026-07-11', '2026-07-16'),

(12, 'Lakshmi', 'Devi', 'Female', '1999-10-05', 'B+', '9876541012', 'lakshmi.devi@gmail.com', 'Hitech City', 'Hyderabad', '2026-07-12', '2026-07

(13, 'Ashok', 'Varma', 'Male', '1990-09-27', 'A+', '9876541013', 'ashok.varma@gmail.com', 'Kompally', 'Hyderabad', '2026-07-13', '2026-07-18'),

(14, 'Pooja', 'Jain', 'Female', '1998-01-15', 'O-', '9876541014', 'pooja.jain@gmail.com', 'Manikonda', 'Hyderabad', '2026-07-14', '2026-07-19'),

(15, 'Harish', 'Kumar', 'Male', '1987-11-08', 'AB+', '9876541015', 'harish.kumar@gmail.com', 'Nallagandla', 'Hyderabad', '2026-07-15', '2026-07-

(16, 'Nikhil', 'Babu', 'Male', '1993-05-21', 'A-', '9876541016', 'nikhil.babu@gmail.com', 'Shamshabad', 'Hyderabad', '2026-07-16', '2026-07-22')

(17, 'Kavya', 'Rani', 'Female', '2001-08-12', 'B+', '9876541017', 'kavya.rani@gmail.com', 'Dilsukhnagar', 'Hyderabad', '2026-07-17', '2026-07-21

(18, 'Sai', 'Teja', 'Male', '1996-04-04', 'O+', '9876541018', 'sai.teja@gmail.com', 'Patancheru', 'Hyderabad', '2026-07-18', '2026-07-23'),

(19, 'Keerthi', 'Naidu', 'Female', '1995-07-29', 'AB-', '9876541019', 'keerthi.naidu@gmail.com', 'Chandanagar', 'Hyderabad', '2026-07-19', '2026

(20, 'Mahesh', 'Rao', 'Male', '1992-02-11', 'B-', '9876541020', 'mahesh.rao@gmail.com', 'Nagole', 'Hyderabad', '2026-07-20', '2026-07-25');


INSERT INTO Rooms
(room_id, room_number, room_type, floor, daily_charge, availability)
VALUES
(1, 'G101', 'General', 1, 1000.00, 'Occupied'),
(2, 'G102', 'General', 1, 1000.00, 'Available'),
(3, 'G103', 'General', 1, 1000.00, 'Occupied'),
(4, 'SP201', 'Semi Private', 2, 2000.00, 'Available'),
(5, 'SP202', 'Semi Private', 2, 2000.00, 'Occupied'),
(6, 'SP203', 'Semi Private', 2, 2000.00, 'Available'),
(7, 'P301', 'Private', 3, 3500.00, 'Occupied'),
(8, 'P302', 'Private', 3, 3500.00, 'Available'),
(9, 'P303', 'Private', 3, 3500.00, 'Occupied'),
(10, 'P304', 'Private', 3, 3500.00, 'Available'),
(11, 'ICU401', 'ICU', 4, 6000.00, 'Occupied'),
(12, 'ICU402', 'ICU', 4, 6000.00, 'Available'),
(13, 'ICU403', 'ICU', 4, 6000.00, 'Occupied'),
(14, 'G104', 'General', 1, 1000.00, 'Available'),
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(15, 'G105', 'General', 1, 1000.00, 'Occupied'),
(16, 'SP204', 'Semi Private', 2, 2000.00, 'Available'),
(17, 'P305', 'Private', 3, 3500.00, 'Occupied'),
(18, 'ICU404', 'ICU', 4, 6000.00, 'Available'),
(19, 'P306', 'Private', 3, 3500.00, 'Occupied'),
(20, 'G106', 'General', 1, 1000.00, 'Available');


INSERT INTO Appointments
(appointment_id, patient_id, doctor_id, appointment_date, appointment_time, visit_type, status, consultation_fee)
VALUES
(1, 1, 1, '2026-07-01', '09:00:00', 'New', 'Completed', 800.00),
(2, 2, 2, '2026-07-02', '09:30:00', 'New', 'Completed', 900.00),
(3, 3, 3, '2026-07-03', '10:00:00', 'Follow-up', 'Completed', 700.00),
(4, 4, 4, '2026-07-04', '10:30:00', 'New', 'Completed', 600.00),
(5, 5, 5, '2026-07-05', '11:00:00', 'New', 'Completed', 750.00),
(6, 6, 6, '2026-07-06', '11:30:00', 'Follow-up', 'Completed', 650.00),
(7, 7, 7, '2026-07-07', '12:00:00', 'New', 'Completed', 500.00),
(8, 8, 8, '2026-07-08', '12:30:00', 'New', 'Completed', 850.00),
(9, 9, 9, '2026-07-09', '01:00:00', 'Follow-up', 'Completed', 950.00),
(10, 10, 10, '2026-07-10', '02:00:00', 'New', 'Completed', 1000.00),
(11, 11, 11, '2026-07-11', '09:15:00', 'New', 'Pending', 900.00),
(12, 12, 12, '2026-07-12', '10:15:00', 'Follow-up', 'Completed', 700.00),
(13, 13, 13, '2026-07-13', '11:15:00', 'New', 'Cancelled', 850.00),
(14, 14, 14, '2026-07-14', '12:15:00', 'New', 'Completed', 950.00),
(15, 15, 15, '2026-07-15', '01:15:00', 'Follow-up', 'Completed', 1000.00),
(16, 16, 16, '2026-07-16', '02:15:00', 'New', 'Completed', 800.00),
(17, 17, 17, '2026-07-17', '03:15:00', 'New', 'Pending', 500.00),
(18, 18, 18, '2026-07-18', '04:15:00', 'Follow-up', 'Completed', 600.00),
(19, 19, 19, '2026-07-19', '05:15:00', 'New', 'Completed', 700.00),
(20, 20, 20, '2026-07-20', '06:15:00', 'New', 'Completed', 650.00);


INSERT INTO Admissions
(admission_id, patient_id, doctor_id, room_id, admission_date, discharge_date, reason)
VALUES
(1, 1, 1, 1, '2026-07-01', '2026-07-05', 'Chest Pain'),
(2, 2, 2, 5, '2026-07-02', '2026-07-04', 'Migraine'),
(3, 3, 3, 7, '2026-07-03', '2026-07-08', 'Leg Fracture'),
(4, 4, 4, 2, '2026-07-04', '2026-07-06', 'High Fever'),
(5, 5, 5, 8, '2026-07-05', '2026-07-10', 'Skin Allergy'),
(6, 6, 6, 3, '2026-07-06', '2026-07-09', 'Ear Infection'),
(7, 7, 7, 9, '2026-07-07', '2026-07-11', 'Diabetes Treatment'),
(8, 8, 8, 4, '2026-07-08', '2026-07-12', 'Pregnancy Checkup'),
(9, 9, 9, 10, '2026-07-09', '2026-07-13', 'Kidney Stones'),
(10, 10, 10, 11, '2026-07-10', '2026-07-14', 'Cancer Therapy'),
(11, 11, 11, 12, '2026-07-11', '2026-07-16', 'Kidney Infection'),
(12, 12, 12, 13, '2026-07-12', '2026-07-17', 'Depression'),
(13, 13, 13, 14, '2026-07-13', '2026-07-18', 'Asthma'),
(14, 14, 14, 15, '2026-07-14', '2026-07-19', 'Stomach Ulcer'),
(15, 15, 15, 16, '2026-07-15', '2026-07-20', 'Thyroid Disorder'),
(16, 16, 16, 17, '2026-07-16', '2026-07-22', 'Eye Surgery'),
(17, 17, 17, 18, '2026-07-17', '2026-07-21', 'Dental Surgery'),
(18, 18, 18, 19, '2026-07-18', '2026-07-23', 'Accident Injury'),
(19, 19, 19, 20, '2026-07-19', '2026-07-24', 'MRI Scan Observation'),
(20, 20, 20, 6, '2026-07-20', '2026-07-25', 'Physiotherapy');

INSERT INTO Medicines
(medicine_id, medicine_name, category, manufacturer, unit_price, stock_quantity, expiry_date)
VALUES
(1, 'Paracetamol 500mg', 'Tablet', 'Sun Pharma', 5.50, 500, '2028-06-30'),
(2, 'Amoxicillin 250mg', 'Capsule', 'Cipla', 12.00, 300, '2028-04-15'),
(3, 'Azithromycin 500mg', 'Tablet', 'Dr. Reddy''s', 25.00, 200, '2028-08-20'),
(4, 'Ibuprofen 400mg', 'Tablet', 'Torrent Pharma', 8.00, 400, '2028-02-10'),
(5, 'Cetirizine 10mg', 'Tablet', 'Cipla', 6.50, 350, '2028-05-18'),
(6, 'Pantoprazole 40mg', 'Tablet', 'Sun Pharma', 15.00, 250, '2028-03-25'),
(7, 'Metformin 500mg', 'Tablet', 'Lupin', 10.00, 450, '2028-09-12'),
(8, 'Amlodipine 5mg', 'Tablet', 'Mankind', 9.50, 300, '2028-07-08'),
(9, 'Atorvastatin 10mg', 'Tablet', 'Zydus', 18.00, 220, '2028-10-14'),
(10, 'Omeprazole 20mg', 'Capsule', 'Abbott', 14.00, 280, '2028-11-30'),
(11, 'Dolo 650', 'Tablet', 'Micro Labs', 4.50, 600, '2028-12-15'),
(12, 'ORS Powder', 'Powder', 'FDC Ltd', 18.00, 180, '2027-12-20'),
(13, 'Insulin Injection', 'Injection', 'Novo Nordisk', 550.00, 80, '2027-11-15'),
(14, 'Vitamin C Syrup', 'Syrup', 'Himalaya', 95.00, 120, '2028-08-28'),
(15, 'Crocin Advance', 'Tablet', 'GSK', 6.00, 500, '2028-09-05'),
(16, 'Diclofenac Gel', 'Ointment', 'Volini', 120.00, 90, '2028-04-18'),
(17, 'Cough Syrup', 'Syrup', 'Benadryl', 110.00, 150, '2028-01-30'),
(18, 'Calcium Tablets', 'Tablet', 'Shelcal', 160.00, 140, '2028-06-12'),
(19, 'Vitamin D3 Capsules', 'Capsule', 'Uprise', 210.00, 100, '2028-07-25'),
(20, 'Saline 500ml', 'Injection', 'Baxter', 85.00, 200, '2029-02-28');

INSERT INTO Prescriptions
(prescription_id, patient_id, doctor_id, medicine_id, dosage, days, prescription_date)
VALUES
(1, 1, 1, 1, '1 Tablet Twice Daily', 5, '2026-07-01'),
(2, 2, 2, 5, '1 Tablet Once Daily', 7, '2026-07-02'),
(3, 3, 3, 4, '1 Tablet Three Times Daily', 10, '2026-07-03'),
(4, 4, 4, 14, '10 ml Twice Daily', 5, '2026-07-04'),
(5, 5, 5, 5, '1 Tablet Once Daily', 14, '2026-07-05'),
(6, 6, 6, 2, '1 Capsule Three Times Daily', 7, '2026-07-06'),
(7, 7, 7, 7, '1 Tablet After Breakfast', 30, '2026-07-07'),
(8, 8, 8, 18, '1 Tablet Daily', 30, '2026-07-08'),
(9, 9, 9, 6, '1 Tablet Before Food', 15, '2026-07-09'),
(10, 10, 10, 13, '10 Units Before Breakfast', 30, '2026-07-10'),
(11, 11, 11, 8, '1 Tablet Daily', 30, '2026-07-11'),
(12, 12, 12, 17, '15 ml Twice Daily', 5, '2026-07-12'),
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(13, 13, 13, 3, '1 Tablet Daily', 5, '2026-07-13'),
(14, 14, 14, 10, '1 Capsule Before Breakfast', 10, '2026-07-14'),
(15, 15, 15, 19, '1 Capsule Weekly', 60, '2026-07-15'),
(16, 16, 16, 16, 'Apply Twice Daily', 14, '2026-07-16'),
(17, 17, 17, 15, '1 Tablet Every 6 Hours', 3, '2026-07-17'),
(18, 18, 18, 20, '500 ml IV Once Daily', 2, '2026-07-18'),
(19, 19, 19, 9, '1 Tablet At Night', 30, '2026-07-19'),
(20, 20, 20, 12, '1 Sachet After Loose Motion', 3, '2026-07-20');


INSERT INTO Bills
(bill_id, patient_id, admission_id, doctor_fee, room_charge, lab_charge, medicine_charge, other_charge, discount, total_amount, payment_status,
VALUES
(1, 1, 1, 800.00, 4000.00, 1200.00, 850.00, 500.00, 350.00, 7000.00, 'Paid', '2026-07-05'),

(2, 2, 2, 900.00, 4000.00, 1800.00, 650.00, 300.00, 450.00, 7200.00, 'Paid', '2026-07-04'),

(3, 3, 3, 1200.00, 17500.00, 3500.00, 1800.00, 1000.00, 1000.00, 24000.00, 'Pending', '2026-07-08'),

(4, 4, 4, 600.00, 2000.00, 900.00, 550.00, 250.00, 300.00, 4000.00, 'Paid', '2026-07-06'),

(5, 5, 5, 750.00, 17500.00, 2200.00, 950.00, 500.00, 900.00, 21000.00, 'Pending', '2026-07-10'),

(6, 6, 6, 650.00, 3000.00, 1100.00, 600.00, 300.00, 350.00, 5300.00, 'Paid', '2026-07-09'),

(7, 7, 7, 500.00, 14000.00, 1500.00, 700.00, 500.00, 700.00, 16500.00, 'Paid', '2026-07-11'),

(8, 8, 8, 850.00, 8000.00, 1300.00, 750.00, 450.00, 350.00, 11000.00, 'Paid', '2026-07-12'),

(9, 9, 9, 950.00, 17500.00, 2600.00, 1100.00, 700.00, 850.00, 22000.00, 'Pending', '2026-07-13'),

(10, 10, 10, 1000.00, 24000.00, 5500.00, 3000.00, 1200.00, 1700.00, 33000.00, 'Paid', '2026-07-14'),

(11, 11, 11, 900.00, 30000.00, 2200.00, 1400.00, 800.00, 1300.00, 34000.00, 'Pending', '2026-07-16'),

(12, 12, 12, 700.00, 30000.00, 1800.00, 900.00, 600.00, 1000.00, 33000.00, 'Paid', '2026-07-17'),

(13, 13, 13, 850.00, 6000.00, 1600.00, 800.00, 350.00, 400.00, 9200.00, 'Paid', '2026-07-18'),

(14, 14, 14, 950.00, 5000.00, 2000.00, 950.00, 450.00, 350.00, 9000.00, 'Pending', '2026-07-19'),

(15, 15, 15, 1000.00, 12000.00, 1800.00, 900.00, 500.00, 700.00, 15500.00, 'Paid', '2026-07-20'),

(16, 16, 16, 800.00, 21000.00, 3500.00, 1600.00, 900.00, 1300.00, 26500.00, 'Pending', '2026-07-22'),

(17, 17, 17, 500.00, 24000.00, 1200.00, 650.00, 300.00, 650.00, 26000.00, 'Paid', '2026-07-21'),

(18, 18, 18, 600.00, 36000.00, 2800.00, 1400.00, 1000.00, 1800.00, 40000.00, 'Pending', '2026-07-23'),

(19, 19, 19, 700.00, 18000.00, 2200.00, 950.00, 600.00, 950.00, 21500.00, 'Paid', '2026-07-24'),

(20, 20, 20, 650.00, 12000.00, 1000.00, 700.00, 350.00, 700.00, 14000.00, 'Paid', '2026-07-25');


INSERT INTO Salary
(salary_id, doctor_id, basic_salary, bonus, deduction, net_salary, salary_month, salary_year, payment_date)
VALUES
(1, 1, 180000.00, 10000.00, 5000.00, 185000.00, 'July', 2026, '2026-07-31'),

(2, 2, 175000.00, 8000.00, 4000.00, 179000.00, 'July', 2026, '2026-07-31'),

(3, 3, 160000.00, 7000.00, 3500.00, 163500.00, 'July', 2026, '2026-07-31'),

(4, 4, 120000.00, 5000.00, 2500.00, 122500.00, 'July', 2026, '2026-07-31'),

(5, 5, 145000.00, 6000.00, 3000.00, 148000.00, 'July', 2026, '2026-07-31'),

(6, 6, 155000.00, 7000.00, 3500.00, 158500.00, 'July', 2026, '2026-07-31'),

(7, 7, 140000.00, 5000.00, 2500.00, 142500.00, 'July', 2026, '2026-07-31'),

(8, 8, 150000.00, 6000.00, 3000.00, 153000.00, 'July', 2026, '2026-07-31'),

(9, 9, 185000.00, 12000.00, 6000.00, 191000.00, 'July', 2026, '2026-07-31'),

(10, 10, 220000.00, 15000.00, 7000.00, 228000.00, 'July', 2026, '2026-07-31'),

(11, 11, 190000.00, 9000.00, 4500.00, 194500.00, 'July', 2026, '2026-07-31'),

(12, 12, 135000.00, 5000.00, 2500.00, 137500.00, 'July', 2026, '2026-07-31'),

(13, 13, 170000.00, 8000.00, 4000.00, 174000.00, 'July', 2026, '2026-07-31'),

(14, 14, 180000.00, 9000.00, 4500.00, 184500.00, 'July', 2026, '2026-07-31'),

(15, 15, 175000.00, 8500.00, 4000.00, 179500.00, 'July', 2026, '2026-07-31'),

(16, 16, 160000.00, 7000.00, 3500.00, 163500.00, 'July', 2026, '2026-07-31'),

(17, 17, 90000.00, 3000.00, 1500.00, 91500.00, 'July', 2026, '2026-07-31'),

(18, 18, 165000.00, 8000.00, 4000.00, 169000.00, 'July', 2026, '2026-07-31'),

(19, 19, 155000.00, 7000.00, 3500.00, 158500.00, 'July', 2026, '2026-07-31'),

(20, 20, 95000.00, 3000.00, 1500.00, 96500.00, 'July', 2026, '2026-07-31');
```

```
CREATE TABLE Laboratory_Tests (
    test_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    test_name VARCHAR(100),
    test_date DATE,
    test_result VARCHAR(50),
    test_cost DECIMAL(10,2),

    FOREIGN KEY (patient_id)
    REFERENCES Patients(patient_id),

    FOREIGN KEY (doctor_id)
    REFERENCES Doctors(doctor_id)
);

INSERT INTO Laboratory_Tests
(test_id, patient_id, doctor_id, test_name, test_date, test_result, test_cost)
VALUES
(1,1,1,'ECG','2026-07-01','Normal',800.00),
(2,2,2,'MRI Brain','2026-07-02','Normal',6500.00),
(3,3,3,'X-Ray Leg','2026-07-03','Fracture',1200.00),
(4,4,4,'Blood Test','2026-07-04','Normal',600.00),
(5,5,5,'Skin Allergy Test','2026-07-05','Positive',1500.00),
(6,6,6,'Hearing Test','2026-07-06','Normal',900.00),
(7,7,7,'Blood Sugar','2026-07-07','High',500.00),
(8,8,8,'Pregnancy Scan','2026-07-08','Normal',2500.00),
(9,9,9,'Kidney Function Test','2026-07-09','Abnormal',1800.00),
(10,10,10,'CT Scan','2026-07-10','Normal',4500.00),
(11,11,11,'Urine Test','2026-07-11','Normal',400.00),
(12,12,12,'Psychological Assessment','2026-07-12','Moderate Stress',2200.00),
(13,13,13,'Pulmonary Function Test','2026-07-13','Abnormal',1700.00),
(14,14,14,'Endoscopy','2026-07-14','Ulcer Detected',3500.00),
(15,15,15,'Thyroid Profile','2026-07-15','Normal',1300.00),
(16,16,16,'Eye Vision Test','2026-07-16','Cataract',700.00),
(17,17,17,'Dental X-Ray','2026-07-17','Cavity Found',900.00),
(18,18,18,'CT Trauma Scan','2026-07-18','Minor Injury',5000.00),
(19,19,19,'MRI Scan','2026-07-19','Normal',6000.00),
(20,20,20,'Physiotherapy Assessment','2026-07-20','Improving',800.00);

CREATE TABLE Hospital_Staff (
    staff_id INT AUTO_INCREMENT PRIMARY KEY,
    staff_name VARCHAR(100) NOT NULL,
    role VARCHAR(50) NOT NULL,
    department_id INT,
    phone VARCHAR(15) UNIQUE,
    email VARCHAR(100) UNIQUE,
    salary DECIMAL(10,2),
    joining_date DATE,
    shift ENUM('Morning','Evening','Night'),

    FOREIGN KEY (department_id)
    REFERENCES Departments(department_id)
);

INSERT INTO Hospital_Staff
(staff_id, staff_name, role, department_id, phone, email, salary, joining_date, shift)
VALUES
(1, 'Lakshmi Priya', 'Nurse', 1, '9123400001', 'lakshmi.priya@hospital.com', 38000.00, '2021-06-10', 'Morning'),
(2, 'Ramesh Naidu', 'Ward Boy', 1, '9123400002', 'ramesh.naidu@hospital.com', 22000.00, '2022-01-15', 'Night'),
(3, 'Anitha Rao', 'Receptionist', 7, '9123400003', 'anitha.rao@hospital.com', 28000.00, '2023-03-12', 'Morning'),
(4, 'Sandeep Kumar', 'Lab Technician', 19, '9123400004', 'sandeep.kumar@hospital.com', 35000.00, '2020-08-20', 'Evening'),
(5, 'Pooja Sharma', 'Pharmacist', 7, '9123400005', 'pooja.sharma@hospital.com', 40000.00, '2021-11-05', 'Morning'),
(6, 'Harish Patel', 'Nurse', 2, '9123400006', 'harish.patel@hospital.com', 39000.00, '2022-04-18', 'Night'),
(7, 'Keerthi Reddy', 'Nurse', 3, '9123400007', 'keerthi.reddy@hospital.com', 38500.00, '2021-09-22', 'Morning'),
(8, 'Vijay Singh', 'Ward Boy', 3, '9123400008', 'vijay.singh@hospital.com', 22500.00, '2023-02-10', 'Evening'),
(9, 'Meena Joseph', 'Nurse', 4, '9123400009', 'meena.joseph@hospital.com', 39500.00, '2020-12-01', 'Morning'),
(10, 'Ravi Teja', 'Lab Technician', 19, '9123400010', 'ravi.teja.staff@hospital.com', 36000.00, '2022-06-14', 'Night'),
(11, 'Divya Menon', 'Receptionist', 7, '9123400011', 'divya.menon.staff@hospital.com', 29000.00, '2023-01-09', 'Evening'),
(12, 'Arun Kumar', 'Pharmacist', 7, '9123400012', 'arun.kumar@hospital.com', 41000.00, '2021-07-19', 'Morning'),
(13, 'Sneha Patel', 'Nurse', 8, '9123400013', 'sneha.patel.staff@hospital.com', 40000.00, '2020-05-25', 'Morning'),
(14, 'Kiran Varma', 'Ward Boy', 10, '9123400014', 'kiran.varma@hospital.com', 23000.00, '2022-10-11', 'Night'),
(15, 'Naveen Reddy', 'Lab Technician', 11, '9123400015', 'naveen.reddy@hospital.com', 35500.00, '2021-03-08', 'Evening'),
(16, 'Bhavana Rao', 'Nurse', 13, '9123400016', 'bhavana.rao@hospital.com', 39200.00, '2022-08-17', 'Morning'),
(17, 'Suresh Babu', 'Pharmacist', 7, '9123400017', 'suresh.babu@hospital.com', 40500.00, '2020-09-30', 'Evening'),
(18, 'Anjali Gupta', 'Receptionist', 18, '9123400018', 'anjali.gupta.staff@hospital.com', 30000.00, '2023-04-21', 'Night'),
(19, 'Rohit Agarwal', 'Nurse', 15, '9123400019', 'rohit.agarwal.staff@hospital.com', 39800.00, '2021-12-13', 'Morning'),
(20, 'Kavya Rani', 'Ward Boy', 18, '9123400020', 'kavya.rani@hospital.com', 23500.00, '2022-07-27', 'Night');

CREATE TABLE Payments (
    payment_id INT AUTO_INCREMENT PRIMARY KEY,
    bill_id INT NOT NULL,
    payment_date DATE NOT NULL,
    payment_method ENUM('Cash','UPI','Credit Card','Debit Card','Net Banking') NOT NULL,
    amount_paid DECIMAL(10,2) NOT NULL,
    transaction_id VARCHAR(50) UNIQUE,
    payment_status ENUM('Paid','Pending','Failed') DEFAULT 'Pending',

    FOREIGN KEY (bill_id)
    REFERENCES Bills(bill_id)
);

INSERT INTO Payments
(payment_id, bill_id, payment_date, payment_method, amount_paid, transaction_id, payment_status)
VALUES
```

```
(1,1,'2026-07-05','UPI',7000.00,'TXN202607050001','Paid'),
(2,2,'2026-07-04','Credit Card',7200.00,'TXN202607040002','Paid'),
(3,3,'2026-07-08','Cash',12000.00,'TXN202607080003','Pending'),
(4,4,'2026-07-06','Debit Card',4000.00,'TXN202607060004','Paid'),
(5,5,'2026-07-10','Net Banking',10000.00,'TXN202607100005','Pending'),
(6,6,'2026-07-09','UPI',5300.00,'TXN202607090006','Paid'),
(7,7,'2026-07-11','Cash',16500.00,'TXN202607110007','Paid'),
(8,8,'2026-07-12','Credit Card',11000.00,'TXN202607120008','Paid'),
(9,9,'2026-07-13','UPI',10000.00,'TXN202607130009','Pending'),
(10,10,'2026-07-14','Net Banking',33000.00,'TXN202607140010','Paid'),
(11,11,'2026-07-16','Debit Card',15000.00,'TXN202607160011','Pending'),
(12,12,'2026-07-17','UPI',33000.00,'TXN202607170012','Paid'),
(13,13,'2026-07-18','Cash',9200.00,'TXN202607180013','Paid'),
(14,14,'2026-07-19','Credit Card',5000.00,'TXN202607190014','Pending'),
(15,15,'2026-07-20','Debit Card',15500.00,'TXN202607200015','Paid'),
(16,16,'2026-07-22','Net Banking',13000.00,'TXN202607220016','Pending'),
(17,17,'2026-07-21','UPI',26000.00,'TXN202607210017','Paid'),
(18,18,'2026-07-23','Credit Card',20000.00,'TXN202607230018','Pending'),
(19,19,'2026-07-24','Cash',21500.00,'TXN202607240019','Paid'),
(20,20,'2026-07-25','Debit Card',14000.00,'TXN202607250020','Paid');

CREATE TABLE Nurses (
    nurse_id INT AUTO_INCREMENT PRIMARY KEY,
    nurse_name VARCHAR(100) NOT NULL,
    gender ENUM('Male','Female','Other') NOT NULL,
    department_id INT NOT NULL,
    qualification VARCHAR(100),
    experience_years INT,
    shift ENUM('Morning','Evening','Night') NOT NULL,
    phone VARCHAR(15) UNIQUE,
    email VARCHAR(100) UNIQUE,

    joining_date DATE,
    salary DECIMAL(10,2),

    FOREIGN KEY (department_id)
    REFERENCES Departments(department_id)
);

INSERT INTO Nurses
(nurse_id, nurse_name, gender, department_id, qualification, experience_years, shift, phone, email, joining_date, salary)
VALUES
(1,'Anitha Reddy','Female',1,'B.Sc Nursing',8,'Morning','9876600001','anitha.reddy@hospital.com','2018-06-12',45000.00),
(2,'Pooja Sharma','Female',2,'GNM',6,'Evening','9876600002','pooja.sharma@hospital.com','2019-04-15',42000.00),
(3,'Ramesh Kumar','Male',3,'B.Sc Nursing',10,'Night','9876600003','ramesh.kumar@hospital.com','2016-08-20',50000.00),
(4,'Sneha Patel','Female',4,'M.Sc Nursing',7,'Morning','9876600004','sneha.patel@hospital.com','2019-01-10',47000.00),
(5,'Kavya Nair','Female',5,'B.Sc Nursing',5,'Evening','9876600005','kavya.nair@hospital.com','2020-09-18',41000.00),
(6,'Harish Rao','Male',6,'GNM',9,'Night','9876600006','harish.rao@hospital.com','2017-05-22',48000.00),
(7,'Divya Gupta','Female',7,'B.Sc Nursing',6,'Morning','9876600007','divya.gupta@hospital.com','2019-11-08',43000.00),
(8,'Lakshmi Devi','Female',8,'M.Sc Nursing',11,'Evening','9876600008','lakshmi.devi@hospital.com','2015-07-14',55000.00),
(9,'Sai Teja','Male',9,'B.Sc Nursing',4,'Night','9876600009','sai.teja@hospital.com','2021-02-28',40000.00),
(10,'Meena Joshi','Female',10,'GNM',12,'Morning','9876600010','meena.joshi@hospital.com','2014-12-05',58000.00),
(11,'Vikram Singh','Male',11,'B.Sc Nursing',8,'Evening','9876600011','vikram.singh@hospital.com','2018-03-25',46000.00),
(12,'Priya Menon','Female',12,'M.Sc Nursing',10,'Night','9876600012','priya.menon@hospital.com','2016-06-30',52000.00),
(13,'Rohit Agarwal','Male',13,'B.Sc Nursing',5,'Morning','9876600013','rohit.agarwal@hospital.com','2020-08-17',42000.00),
(14,'Neha Jain','Female',14,'GNM',7,'Evening','9876600014','neha.jain@hospital.com','2019-10-21',44000.00),
(15,'Arjun Reddy','Male',15,'B.Sc Nursing',9,'Night','9876600015','arjun.reddy@hospital.com','2017-01-19',49000.00),
(16,'Keerthi Rao','Female',16,'M.Sc Nursing',6,'Morning','9876600016','keerthi.rao@hospital.com','2020-04-09',45000.00),
(17,'Sunil Babu','Male',17,'GNM',8,'Evening','9876600017','sunil.babu@hospital.com','2018-09-13',46000.00),
(18,'Bhavani Devi','Female',18,'B.Sc Nursing',11,'Night','9876600018','bhavani.devi@hospital.com','2015-11-27',56000.00),
(19,'Mahesh Varma','Male',19,'M.Sc Nursing',10,'Morning','9876600019','mahesh.varma@hospital.com','2016-02-11',53000.00),
(20,'Swathi Reddy','Female',20,'B.Sc Nursing',5,'Evening','9876600020','swathi.reddy@hospital.com','2021-01-05',41000.00);



CREATE TABLE Insurance (
    insurance_id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT NOT NULL,
    company_name VARCHAR(100) NOT NULL,
    policy_number VARCHAR(30) UNIQUE NOT NULL,
    coverage_amount DECIMAL(12,2) NOT NULL,
    claim_amount DECIMAL(12,2) DEFAULT 0,
    claim_status ENUM('Approved','Pending','Rejected') DEFAULT 'Pending',
    policy_start_date DATE,
    policy_end_date DATE,

    FOREIGN KEY (patient_id)
    REFERENCES Patients(patient_id)
);


INSERT INTO Insurance
(insurance_id, patient_id, company_name, policy_number, coverage_amount, claim_amount, claim_status, policy_start_date, policy_end_date)
VALUES
(1,1,'Star Health','SH100001',500000.00,7000.00,'Approved','2026-01-01','2026-12-31'),
(2,2,'Niva Bupa','NB100002',300000.00,7200.00,'Approved','2026-01-15','2027-01-14'),
(3,3,'ICICI Lombard','IL100003',1000000.00,24000.00,'Pending','2026-02-01','2027-01-31'),
(4,4,'HDFC ERGO','HE100004',250000.00,4000.00,'Approved','2026-01-10','2027-01-09'),
(5,5,'Care Health','CH100005',400000.00,21000.00,'Pending','2026-03-01','2027-02-28'),
(6,6,'Aditya Birla Health','AB100006',350000.00,5300.00,'Approved','2026-02-15','2027-02-14'),
(7,7,'Star Health','SH100007',500000.00,16500.00,'Approved','2026-01-20','2027-01-19'),
(8,8,'Niva Bupa','NB100008',300000.00,11000.00,'Approved','2026-03-10','2027-03-09'),
(9,9,'ICICI Lombard','IL100009',800000.00,22000.00,'Pending','2026-04-01','2027-03-31'),
(10,10,'HDFC ERGO','HE100010',1200000.00,33000.00,'Approved','2026-01-05','2027-01-04'),
(11,11,'Care Health','CH100011',450000.00,34000.00,'Pending','2026-02-20','2027-02-19'),
(12,12,'Aditya Birla Health','AB100012',350000.00,33000.00,'Approved','2026-03-15','2027-03-14'),
```

```
(13,13,'Star Health','SH100013',500000.00,9200.00,'Approved','2026-01-18','2027-01-17'),
(14,14,'Niva Bupa','NB100014',250000.00,9000.00,'Pending','2026-04-10','2027-04-09'),
(15,15,'ICICI Lombard','IL100015',600000.00,15500.00,'Approved','2026-05-01','2027-04-30'),
(16,16,'HDFC ERGO','HE100016',900000.00,26500.00,'Pending','2026-03-20','2027-03-19'),
(17,17,'Care Health','CH100017',300000.00,26000.00,'Approved','2026-02-05','2027-02-04'),
(18,18,'Aditya Birla Health','AB100018',700000.00,40000.00,'Pending','2026-01-25','2027-01-24'),
(19,19,'Star Health','SH100019',500000.00,21500.00,'Approved','2026-04-15','2027-04-14'),
(20,20,'Niva Bupa','NB100020',350000.00,14000.00,'Approved','2026-05-10','2027-05-09');
```

project questions.sql

```
SELECT * FROM hospital_management_system.bills;
use hospital_management_system;

-------------1. Find the Top 5 Highest Paid Doctors---------
-----Concepts: ORDER BY, LIMIT-------


SELECT d.doctor_name,d.doctor_id,
       dep.department_name,
       s.net_salary
FROM Doctors d
JOIN Salary s
    ON d.doctor_id = s.doctor_id
JOIN Departments dep
    ON d.department_id = dep.department_id
ORDER BY s.net_salary DESC
LIMIT 5;


-------2. Department-wise Revenue-------
------Concepts: JOIN, GROUP BY, SUM---------


SELECT dep.department_name,
       SUM(b.total_amount) AS Total_Revenue
FROM Bills b
JOIN Admissions a
    ON b.admission_id = a.admission_id
JOIN Doctors d
    ON a.doctor_id = d.doctor_id
JOIN Departments dep
    ON d.department_id = dep.department_id
GROUP BY dep.department_name
ORDER BY Total_Revenue DESC;


-------3. Doctors Earning Above Average Salary----------------------
----------Concepts: Subquery-----------------


SELECT d.doctor_name,d.doctor_id,
       s.net_salary
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id
WHERE s.net_salary >
(
SELECT AVG(net_salary)
FROM Salary
);


-------4. Patients with Pending Bills-------
---------Concepts: JOIN + Filter-------

SELECT p.first_name,
       p.last_name,
       b.total_amount
FROM Patients p
JOIN Bills b
ON p.patient_id=b.patient_id
WHERE b.payment_status='Pending';



-------5. Top 5 Revenue Generating Patients-------


SELECT p.first_name,p.patient_id,
       p.last_name,
       SUM(b.total_amount) Total_Bill
FROM Patients p
JOIN Bills b
ON p.patient_id=b.patient_id
GROUP BY p.patient_id,p.first_name,p.last_name
ORDER BY Total_Bill DESC
LIMIT 5;

------------6. Rank Doctors by Salary (RANK)---------------------

SELECT
    d.doctor_name,d.doctor_id,
    dep.department_name,
    s.net_salary,
    RANK() OVER(ORDER BY s.net_salary DESC) AS Salary_Rank
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id
JOIN Departments dep
ON d.department_id=dep.department_id;

------------------7. Dense Rank Doctors by Salary----------------------
```

```
SELECT
    d.doctor_name,d.doctor_id,
    s.net_salary,
    DENSE_RANK() OVER(ORDER BY s.net_salary DESC) AS Salary_Rank
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id;


------------------8. Rank Patients by Bill Amount-------------------

SELECT
    p.first_name,p.patient_Id,
    p.last_name,
    b.total_amount,
    ROW_NUMBER() OVER(ORDER BY b.total_amount DESC) AS Bill_Rank
FROM Patients p
JOIN Bills b
ON p.patient_id=b.patient_id;


---------------9. Previous Doctor Salary (LAG)-----------------

SELECT
    d.doctor_name,d.doctor_id,
    s.net_salary,
    LAG(s.net_salary)
    OVER(ORDER BY s.net_salary) AS Previous_Salary
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id;

--------------------10. Next Doctor Salary (LEAD)------------------

SELECT
    d.doctor_name,d.doctor_id,
    s.net_salary,
    LEAD(s.net_salary)
    OVER(ORDER BY s.net_salary) AS Next_Salary
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id;


-----------------11. Running Hospital Revenue-----------------

SELECT
    bill_date,
    total_amount,
    SUM(total_amount)
    OVER(ORDER BY bill_date) AS Running_Revenue
FROM Bills;


-------------------12. Monthly Revenue Trend----------------

SELECT
    MONTH(bill_date) AS Month,
    SUM(total_amount) AS Revenue
FROM Bills
GROUP BY MONTH(bill_date)
ORDER BY Month;

-----------13. Average Bill by Department----------------

select
    dep.department_name,
    ROUND(AVG(b.total_amount),2) AS Average_Bill
FROM Bills b
JOIN Admissions a
ON b.admission_id=a.admission_id
JOIN Doctors d
ON a.doctor_id=d.doctor_id
JOIN Departments dep
ON d.department_id=dep.department_id
GROUP BY dep.department_name;


-------------14. Department Generating Highest Revenue-----------------

SELECT
    dep.department_name,
    SUM(b.total_amount) Revenue
FROM Bills b
JOIN Admissions a
ON b.admission_id=a.admission_id
JOIN Doctors d
ON a.doctor_id=d.doctor_id
JOIN Departments dep
ON d.department_id=dep.department_id
GROUP BY dep.department_name
ORDER BY Revenue DESC
LIMIT 1;

----------------15. Doctor Treating Maximum Patients-------
```

```
SELECT
    d.doctor_name,
    COUNT(a.patient_id) AS Total_Patients
FROM Doctors d
JOIN Appointments a
ON d.doctor_id=a.doctor_id
GROUP BY d.doctor_id,d.doctor_name
ORDER BY Total_Patients DESC
LIMIT 1;


--------------16. Most Frequently Prescribed Medicine--------------------

SELECT
    m.medicine_name,
    COUNT(*) AS Times_Prescribed
FROM Prescriptions p
JOIN Medicines m
ON p.medicine_id=m.medicine_id
GROUP BY m.medicine_name
ORDER BY Times_Prescribed DESC
LIMIT 1;

-----------------17. Patients Admitted More Than Once-----------------

SELECT
    p.first_name,
    p.last_name,
    COUNT(a.admission_id) AS Admissions
FROM Patients p
JOIN Admissions a
ON p.patient_id=a.patient_id
GROUP BY p.patient_id,p.first_name,p.last_name
HAVING COUNT(*)>1;



-------------18. Average Hospital Stay-------------------


SELECT
ROUND(AVG(DATEDIFF(discharge_date,admission_date)),2)
AS Average_Stay
FROM Admissions;

---------19. Longest Hospital Stay-------------------

SELECT

p.patient_id,
p.first_name,
p.last_name,
DATEDIFF(a.discharge_date,a.admission_date)
AS Stay_Days
FROM Admissions a
JOIN Patients p
ON a.patient_id=p.patient_id
ORDER BY Stay_Days DESC
LIMIT 1;

-------------------20. Insurance Company with Highest Claims-----------------

SELECT
company_name,
SUM(claim_amount) AS Total_Claims
FROM Insurance
GROUP BY company_name
ORDER BY Total_Claims DESC
LIMIT 1;

------------21. Room Occupancy-------------------

SELECT
r.room_type,
COUNT(a.room_id) AS Occupied
FROM Rooms r
LEFT JOIN Admissions a
ON r.room_id=a.room_id
GROUP BY r.room_type;

----------------22. Doctor-wise Revenue---------------------

SELECT
d.doctor_name,
d.doctor_id,
SUM(b.total_amount) AS Revenue
FROM Doctors d
JOIN Admissions a
ON d.doctor_id=a.doctor_id
JOIN Bills b
ON a.admission_id=b.admission_id
GROUP BY d.doctor_id
ORDER BY Revenue DESC;

----------------23. Department-wise Average Salary-----------

SELECT
```

```
dep.department_name,
ROUND(AVG(s.net_salary),2) Average_Salary
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id
JOIN Departments dep
ON d.department_id=dep.department_id
GROUP BY dep.department_name;

--------------24. Top 3 Doctors in Every Department-------------------

SELECT *
FROM
(
SELECT
dep.department_name,
d.doctor_name,
s.net_salary,
ROW_NUMBER() OVER
(
PARTITION BY dep.department_name
ORDER BY s.net_salary DESC
) rn
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id
JOIN Departments dep
ON d.department_id=dep.department_id
)t
WHERE rn<=3;-

------------------25. Doctors Earning More Than Department Average (Correlated Subquery)--------------

SELECT
d.doctor_name,
s.net_salary
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id
WHERE s.net_salary >
(
SELECT AVG(s2.net_salary)
FROM Doctors d2
JOIN Salary s2
ON d2.doctor_id=s2.doctor_id
WHERE d2.department_id=d.department_id
);

-------------26. Patients Having Insurance and Pending Bills (EXISTS)------------------

SELECT
p.patient_id,
p.first_name,
p.last_name
FROM Patients p
WHERE EXISTS
(
SELECT 1
FROM Insurance i
WHERE i.patient_id=p.patient_id
)
AND EXISTS
(
SELECT 1
FROM Bills b
WHERE b.patient_id=p.patient_id
AND b.payment_status='Pending'
);

-------------------27. Medicines Never Prescribed (NOT EXISTS)---------------

SELECT

m.medicine_name
FROM Medicines m
WHERE NOT EXISTS
(
SELECT 1
FROM Prescriptions p
WHERE p.medicine_id=m.medicine_id
);

---------------28. Top Revenue Department (CTE)----------------------

WITH RevenueCTE AS
(
SELECT
dep.department_name,
SUM(b.total_amount) Revenue
FROM Bills b
JOIN Admissions a
ON b.admission_id=a.admission_id
JOIN Doctors d
ON a.doctor_id=d.doctor_id
JOIN Departments dep
ON d.department_id=dep.department_id
```

```
GROUP BY dep.department_name
)

SELECT *
FROM RevenueCTE
ORDER BY Revenue DESC
LIMIT 1;

-------------29. Doctors Paid More Than ALL Doctors in Department 1 (ALL)-------------------

SELECT
d.doctor_id,
d.doctor_name,
s.net_salary
FROM Doctors d
JOIN Salary s
ON d.doctor_id=s.doctor_id
WHERE s.net_salary > ALL
(
SELECT s2.net_salary
FROM Doctors d2
JOIN Salary s2
ON d2.doctor_id=s2.doctor_id
WHERE d2.department_id=1
);

----------30. Complete Patient Treatment Report (8+ JOINs)-------------------

SELECT
p.first_name,
p.last_name,
d.doctor_name,
dep.department_name,
r.room_number,
lt.test_name,
m.medicine_name,
b.total_amount,
pay.payment_status,
i.company_name
FROM Patients p
JOIN Admissions a
    ON p.patient_id=a.patient_id
JOIN Doctors d
    ON a.doctor_id=d.doctor_id
JOIN Departments dep
    ON d.department_id=dep.department_id
JOIN Rooms r
    ON a.room_id=r.room_id
LEFT JOIN Laboratory_Tests lt
    ON p.patient_id=lt.patient_id
LEFT JOIN Prescriptions pr
    ON p.patient_id=pr.patient_id
LEFT JOIN Medicines m
    ON pr.medicine_id=m.medicine_id
LEFT JOIN Bills b
    ON a.admission_id=b.admission_id
LEFT JOIN Payments pay
    ON b.bill_id=pay.bill_id
LEFT JOIN Insurance i
    ON p.patient_id=i.patient_id
ORDER BY p.patient_id;
```